ANNUAL PROGRESS REPORT: 2023-24

Name of centre: AICRP on Maize, Mandya (UAS, Bangalore)

I. ACHIEVEMENTS SUMMARY :

A. Maize crop Improvement:

- The hybrid MAH 14-138 was recommended for release in 41st State Seeds Sub Committee meeting held on 23.01.2024.
- The hybrid MAH 15-84 was recommended for release in Zone 5 & 6 respective ZREP meeting held during April, 2024.
- Totally 22 co-ordinated trials, 14 station trials, 2 Multi location trials, 9 station breeding programs, 2 Inter institutional collaborative programs, 2 PG research supported programs and 65 farm trials were conducted under crop improvement.
- Induced double haploid for the promising hybrids and generated 1117 DH populations for evaluation.
- Heterotic grouping of inbreds of normal and specialty maize was initiated with objective to identify diverse and contrasting parents for hybrid development. Totally 76 (normal), 51 (popcorn and 85 (sweet corn) inbreds were used for heterotic grouping.
- Popularized the new hybrids (MAH 14-5 and MAH 14-138) developed by the university through Front line demonstration in 125 acres in Maize growing regions of Mandya, Mysore, Hassan and Chamarajanagara districts during *Kharif*-2023 and 50 acres SCSP program implemented in maize growing area Sollepura Haadi-Chakkodanahalli, H.D.Kote Tq., Mysuru Dist.during Rabi 2023-24.

B. Maize Pathology: SDM

- A technology to manage the Maydis Leaf Blight of maize caused by *Bipolaris maydis* has been recommended for Zone 6 during the year 2023-24.
- Developed the weather based forecasting model to predict the Maize Downy Mildew disease using Machine Learning algorithms and aaccessible web app at https://mdmpdi.pythonanywhere.com/, demonstrated advanced forecasting for sustainable agriculture.
- Identified the 15 stable maize inbreds resistant to *Peronosclerospora sorghi* causing the Sorghum Downy Mildew Disease.
- Liquid Chromatography-Mass Spectrometry analysis in both resistant and susceptible genotypes, highlighting the presence of essential phenolic and flavonoid compounds. The analysis identified 18 distinct phenolic compounds, among which Salicylic acid, p-coumaric acid, o-coumaric acid, caffeic acid, and ferulic acid displayed significant fold increases in resistant genotypes. Additionally, flavonoid profiling revealed 15 compounds, showcasing notable elevations, especially in naringenin, catechin, and epicatechin levels, in resistant genotypes. These findings highlight the plant's chemical defence responsiveness, bolstering resilience against *Peronosclerospora sorghi*.

C. Maize Pathology: TLB

- Evaluation of Maize genotypes against Turcicum Leaf Blight incited by *Exserohilum turcicum during Kharif* and *Rabi*:
  A total of 14+3 trials were conducted during *Kharif* 2023 and *Rabi* 2022-23 and promising entries which recorded ≤ 3 scale (1-9 scale) are as follows:
  ***Kharif* 2023:** BH418103, CT2011312, GH20113, GH20115, HARLAL1001, HT523069, IM38619, KMH23-14, NMH4790, PA6354, PA6401, PM23114, PM23117, SMSX22300, SRMH66M33, CP988, HARLAL24, IQ8701, KMH8206, SYN221610L, KMH8333, CMH8-287, CP858 and HT 5109.
  Rabi 2022-23 AHD8722, MMH22-62, PM222041, PM22204L, ADV7559, PM21207L, PM20205L, NMH713 (Ch), P3522 (Ch) and SURYA (Ch).

- In maize disease trap nursery, ten inbred lines were sown for observation on natural occurrence of maize diseases. Turcicum Leaf Bilght is a major disease at ZARS with maximum disease recorded 9.0 on CM 400, CML 500 and BML 6. MLB was noticed on CM 600, Surya and LM 14. SDM was noticed on CM 500 and Surya. CLS was noticed on BML 6 and IIMR SBT POOL.

- Screening of maize lines resistant to TLB disease: Out of 50 promising genotypes screened, only 12 genotypes namely NAI-137, NAI-175, NAI-187, NAI-204, NAI-212, NAI-217, MAI-2, MAI-264, MAI-365,CML-300, QMSC-36 and SKV-50 recorded resistant reaction with less than 3 grade (on 1-9 scale).

- Pest and Disease survey and surveillance in different maize growing areas of Karnataka. The major diseases like Sorghum Downey mildew (55%), TLB (70%), MLB (45%), CLS (20%0), P.Rust(40%) and Post Flowering Stalk Rots (PFSR)(25%) were noticed. Severe intensity of TLB was recorded in almost all the field surveyed. Maximum disease incidence noticed in Hassan district (85%) followed by Mysore (75%), Chamarajanagara (70%) and Mandya (65%) districts.

- In management of polysora rust disease, among all tested fungicides the best treatment was the treatment (T4) receiving Foliar spray of Azoxystrobin 18.2% +Difenconazole 11.4 % @ 1 ml/litre of water at 35 and 50 DAS recorded lowest PDI (**28.13**), maximum per cent disease control (**62.6%**) and higher grain yield (**82.6** q/ha) when compared to control which recorded higher PDI (**75.40**) and lower grain yield (**25.4** q/ha).

- In the racial pattern studies, highest disease severity was observed in **Hassan, Mandya and Shivamogga** districts and they are identified as hot spot locations for TLB. Based on the **sporulation, lesion length, latent period, per cent disease index and virulence index, Hassan (Et2) and Mandya (Et1) isolates** were found more aggressive and virulent than the remaining isolates. These virulent isolates were used for screening of maize inbred lines for disease reaction under artificial epiphytotic conditions in National programme.

- In the experiment on assessment of yield loss due to TLB, the results revealed that the maximum yield loss upto the tune of 40.4% was recorded.

D. Maize Nutrition:

- Among the 16 Popcorn genotypes tested the average popping percentage varied between 86.40 to 98.36%. PP 09 recorded higher popping traits as compared to other entries and checks.
- Among the 16 Popcorn genotypes tested the average popping percentage varied between 86.36 to 96.23%. MPC-2141 has got highest popping percentage (96.23%) followed by IMHP-2101 (95.95%) and APCH-2 (95.67%).
- The nutritional composition of pasta revealed that HQPM pasta, and IQPMH Pasta contains high protein (12.69 & 13.48g), crude fiber (12.96 &12.09), iron (3.53& 3.25mg) and calcium (38.06& 37.76mg) contents.
- The ready to cook (RTC) Nutri maize (QPM) HQPM-5 based pastas standardized using HQPM-5 maize flour, HQPM-5 maize suji Black gram flour and Gum. The ratio of 60:40 was acceptable by the sensory evaluation panelists. The nutritional composition of pasta revealed that HQPM pasta, and IQPMH Pasta contains high protein (12.69 & 13.48g), crude fiber (12.96 &12.09), iron (3.53& 3.25mg) and calcium (38.06& 37.76mg) contents.
- In the storage studies HQPM-5 Pasta stored in MPP packaging has been found acceptable up to 6 months as compared to HDPE packaging based on sensory and biochemical evaluations.
- The nutritional composition of pasta revealed that Foxtail millet pasta, and little millet pasta contains high protein (15.41 & 11.82), crude fiber (14.53 &13.89), iron (2.61& 2.59mg) and calcium (9.93 & 9.63 mg) contents. The foxtail millet Pasta stored in MPP packaging has been found acceptable up to 6 months as compared to HDPE packaging based on sensory and biochemical evaluations.
- The little millet Pasta stored in MPP packaging has been found acceptable up to 6 months as compared to HDPE packaging based on sensory and biochemical evaluations.
- The ready to cook (RTC) little millet dosa standardized using little millet flour black gram dhal, sprouted green gram, bengal gram dhal, rice flakes, fenugreek seeds. The ratio of 60:40 was acceptable by the sensory evaluation panelists.
- The ready to cook (RTC) foxtail millet Idli standardized using foxtail millet suji black gram dhal, sported green gram, black gram dhal, rice flakes, fenugreek seeds, green chilly, carrot and shepu leaf powder. The ratio of 60:40 was acceptable by the sensory evaluation panelists.
- The nutritional composition of Nutri bars revealed that HQPM-5 NB and IQPMHNB contain high protein (8.66 & 8.68g), crude fiber (6.38 & 6.35g), and Fat (9.12 & 9.11g) contents.
- The nutritional composition of Nutri laddus revealed that QPM Laddu (Finger millet) and QPM Laddu (Little millet) contain high protein (15.45 & 17.31g), crude fiber (4.67 & 4.00g), and Fat (15.33 & 17.94g) contents.
- The ready to cook (RTC) B-Carotene enriched RTC Maize suji Carrot halwa mix (Jaggery) standardized using Maize Suji, Dehydrated Carrot powder, Jaggery and other ingredients. The ratio of 60:40 was acceptable by the sensory evaluation panelists.
- The ready to cook (RTC) B-Carotene enriched RTC Maize suji Pumpkin halwa mix (Sugar) standardized using Maize Suji, Dehydrated Pumpkin powder, Sugar and other ingredients. The ratio of 60:40 was acceptable by the sensory evaluation panelists.

DETAIL REPORT

A. General Information

Year of establishment: 1981

Season-wise area, production and productivity of crop in the state (5 years) in Karnataka State:

Year	Area (Lakh ha)		Total (Lakh ha)	Production (Lakh tones)		Total (Lakh tones)	Productivity (kg/ha)		Total (kg/ha)
	Kharif	*Rabi*		*Kharif*	*Rabi*		*Kharif*	*Rabi*	
2019-20	13.55	1.87	**13.97**	43.66	2.95	**47.68**	3392	3569	**3414**
2020-21	15.32	1.49	**17.27**	56.83	5.40	**63.72**	3709	3691	**3689**
2021-22	15.11	0.97	**16.42**	48.82	3.31	**53.38**	3231	3415	**3251**
2022-23	14.33	1.57	**15.91**	47.99	4.61	**52.60**	3347	2936	**3347**
2023-24*	17.43	1.68	**19.40**	47.49	6.41	**54.91**	2724	3825	**2830**

*Third advance estimates data received from KSDA, GOK, Bangalore.

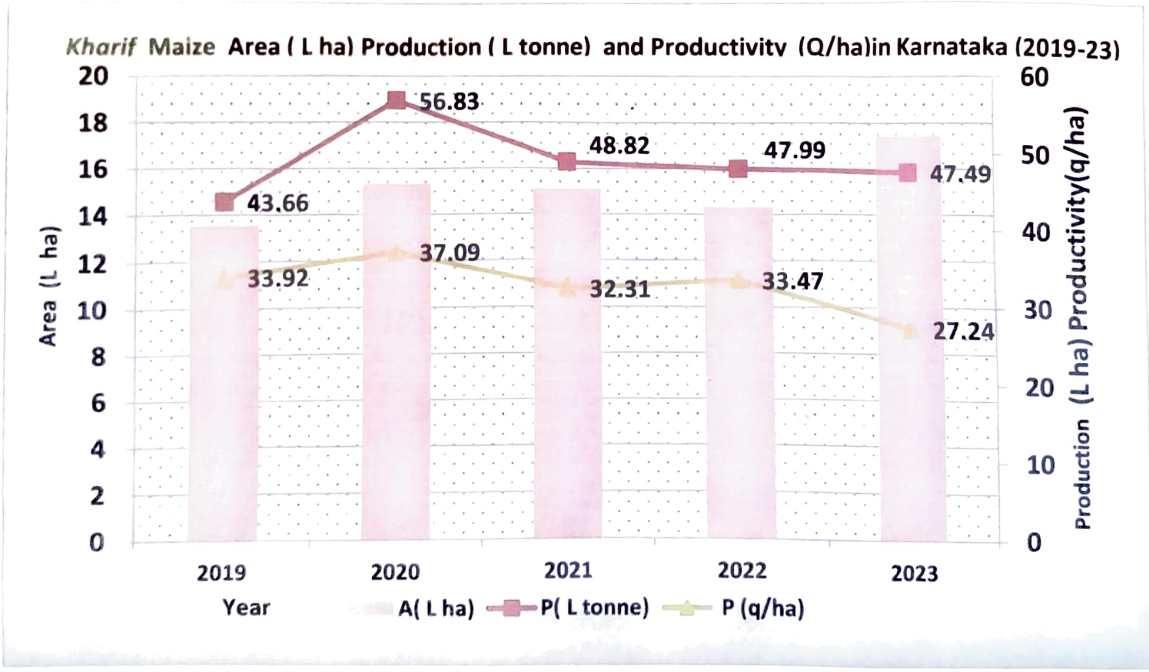


Fig 1: Kharif Maize area, Production and productivity during last five year (2019-2023) in Karnataka

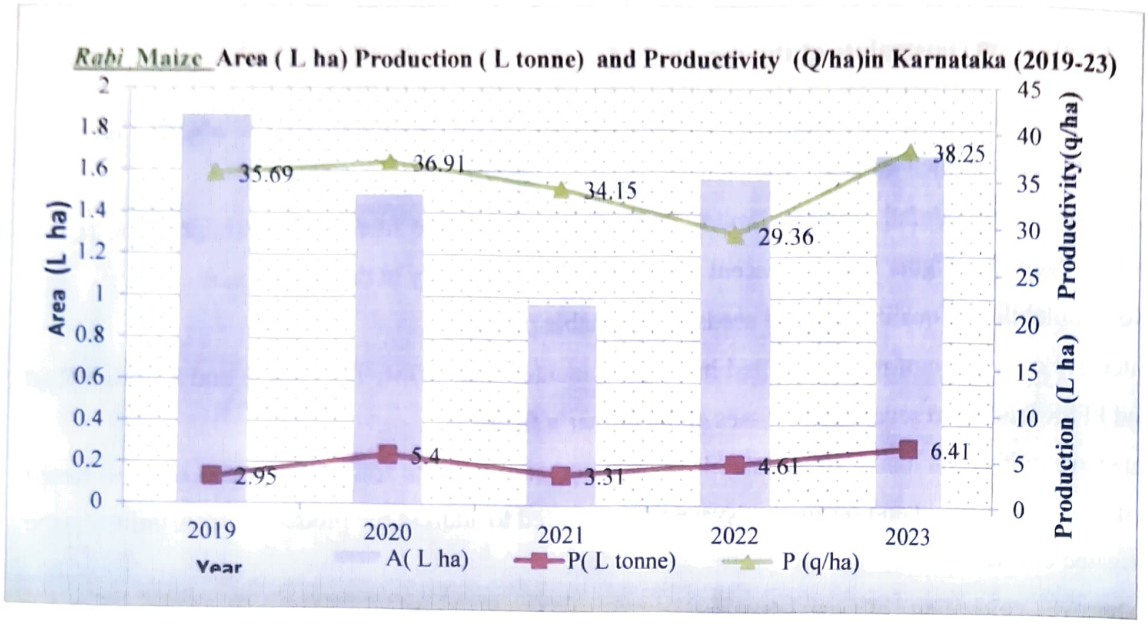


Fig 2 : Rabi Maize area, Production and productivity during last five year(2019-2023) in Karnataka

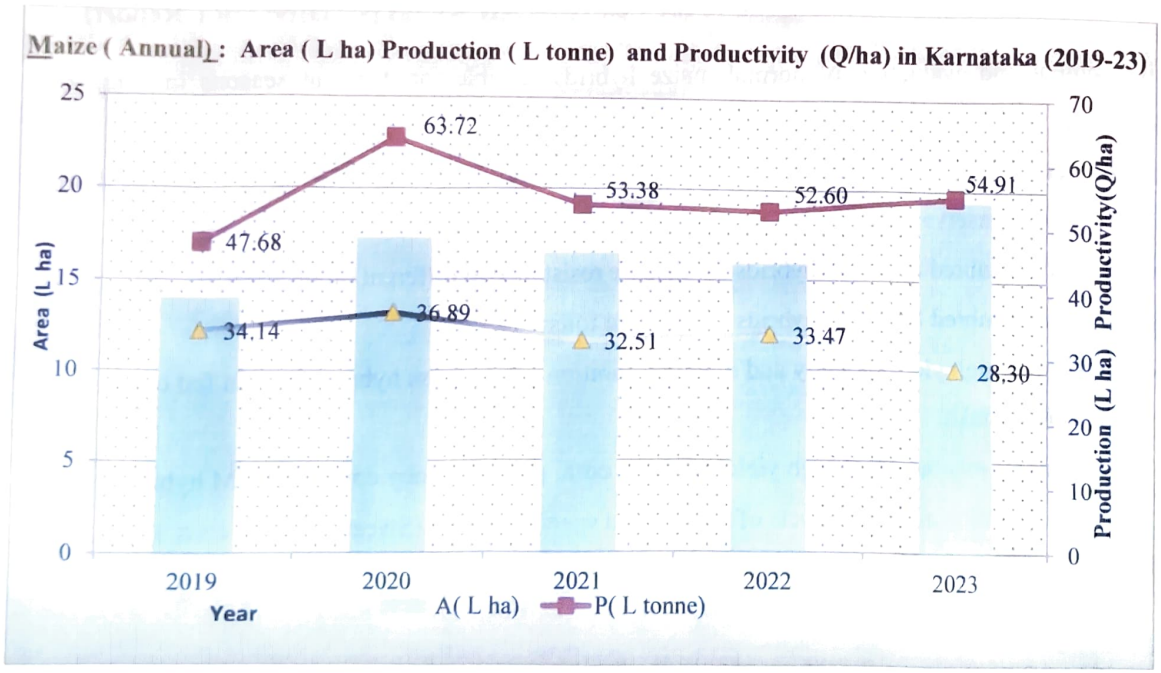


Fig 3 : Annual Maize area, Production and productivity during last five year(2019-2023)in Karnataka

Agro-ecological region for which the centre is working: Peninsular Zone (PZ)

B. Production constraints of the region:

- Maize is grown both under irrigated (30%) and rain fed (70 %) conditions with variable soil and agro climatic factors which influence the yield.
- Drought /erratic rainfall create an intermittent stress and become foremost constraint in the region which resulted in higher losses in recent years and more pertinently in the current year.
- Non availability of quality certified seeds at reasonable prices.
- Intensive cultivation of maize resulted in frequent incidence of SDM, TLB, MLB and Polysora Rust and PFSR leading to severe yield losses at the farmer's field.
- Intensive cultivation maize after maize leading to imbalance in the soil nutrients and soil test based fertilizer management and fertigation system are required to address the production constraints in the irrigated cultivation.
- Extensive cultivation of specialty maize like Baby corn, Popcorn and Sweet corn need standardization of spacing nutrients and agronomic management practices to increase the productivity and quality with enhanced shelf life.

C. Thrust areas of research assigned to the Centre by AICRP (as per latest QRT Report)

- Development and evaluation of normal maize hybrids suitable for different seasons in southern Karnataka.
- Development of maize inbreds resistant to TLB, Polysora rust and PFSR and maintained for utilization and conservation.
- Development of inbred lines and hybrids for disease resistance to different biotic stresses.
- Development of inbred lines and hybrids for drought tolerance.
- Development of high yielding early and medium duration single cross hybrids for rain fed ecosystem in southern Karnataka.
- Development and evaluation of high yielding sweet corn, popcorn, baby corn and QPM hybrids.
- Development of value added products of normal and specialty corns (Sweet corn, popcorn, baby corn and QPM) and their commercialization.

D. WEATHER CONDITIONS

The weather conditions at Mandya center reported from Jan 2023 to Dec -2023 recorded actual rainfall of 524.4 mm with a deficit of 26.4 % over the normal rainfall (712.2 mm). The rainfall in the months of April and May recorded higher rainfall over the normal remaining months recorded below normal. The number of rainy days also is recorded less as compared to the normal.

Tabli-2: Monthly mean weather parameters at ZARS, V.C. Farm, Mandya Centre for the year -2023

MONTH	MAX (^{0}C)	MIN (^{0}C)	RH I (%)	RH II (%)	WS (Km)	SS H (hrs)	Actual rainfall (mm)	Normal rainfall (mm)	% Dev	Actual R.Days	Normal R.Days
JAN	30.1	14.7	82	62	2.2	8.7	0	1.7	-100	0	0
FEB	32.3	15.4	81	64	2.7	7.8	0	4.4	-100	0	0
MAR	33.7	18.2	72	57	2.7	8.5	13.0	13.4	-3.0	1	0
APR	35.5	21.2	75	54	1.8	9.1	68.0	49.5	37.4	4	2
MAY	33.1	22.2	78	56	2.4	8.9	163.0	96.5	76.0	6	7
JUN	31.6	21.9	79	57	4.7	5.6	1.0	63.0	-98.3	0	3
JUL	28.6	19.5	84	68	6.7	0.0	40.0	53.9	-22.9	5	2
AUG	32.3	19.4	85	59	4.7	0.5	45.7	83.7	-39.9	2	13
SEP	28.2	20.6	85	59	4.3	6.7	65.6	133.1	-50.9	6	6
OCT	31.3	19.4	86	60	1.9	7.1	89.5	145.8	-36.6	4	15
NOV	29.7	19.4	83	59	2.0	6.4	56.6	52.6	6.4	5	1
DEC	29.8	18.0	82	58	2.1	6.5	0	14.6	-100	0	0
Total							**524.4**	**712.2**	**-26.4**	**33**	**49**

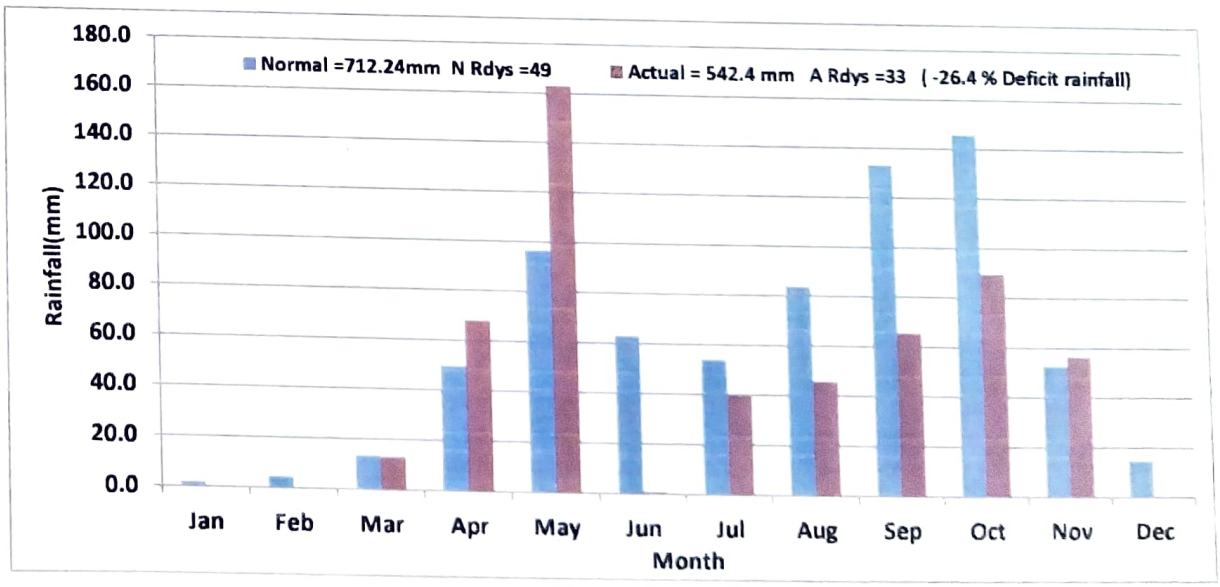


Fig 1.Monthly rainfall during 2023 and normal rainfall at V.C Farm, Mandya.

Maize crop season wise scenario in the Karnataka during the year 2023

Karnataka is one of the main maize producing states having highest area and production contributing in 17% of total area and production of the country. Majority of maize is grown under rainfed (70%) and irrigated (30%) with majority in Kharif (80%) followed by Rabi (15%) and 5% in Summer seasons. Normally yellow / orange grain maize is grown mainly for feed (60%), food (17%) and pharmaceutical and industrial purposes (13%) in the state. Specialty corn like sweet corn, popcorn and baby corn are also grown considerably in an area of 10,000-12,000 ha covering 6 to 8% of the total maize area in the state.

During the year 2023-24, South Interior Karnataka recorded 27, 31 and 80% deficit rainfall during Kharif 2023, Rabi 2023 and summer 2024 respectively. Further the state as whole received 25, 38, 34% deficit rainfall respectively. Hence, there was considerable reduction in the productivity during both Kharif and Rabi seasons due to severe drought and adverse weather conditions. More than 78.2% maize area was affected by drought and intermittent stress conditions during Kharif season and severe moisture and heat stresses during Rabi season also expecting proportionate reduction.

District wise maize area affected due to drought during kharif 2023

Sl. No.	District	Normal kharif area (Ha)	Major maize area affected (ha)
1	Haveri	223065	212181
2	Vijayanagara	180000	200295
3	Koppal	93373	128675
4	Gadag	90341	124055
5	Davanagere	139302	120034
6	Vijayapura	79089	81761
7	Hassan	122460	76434
8	Chitradurga	114616	68286
9	Dharwad	64099	65940
10	Belagavi	226126	61663
11	Shivamogga	65541	38241
12	Bagalkote	79297	36820
13	Chikkaballapur	55154	33005
14	Mysuru	49602	32261
15	Tumakuru	33534	18920
16	Ballari	18606	17390
17	Chamarajanagara	38045	12489
18	Uttarkannada	10312	7801
19	Chikkamangaluru	21262	6476
20	Bengaluru Rural	8804	3478
21	Kodagu	408	2170
22	Mandya	6233	1245
State Total		**1719269**	**1350668**

The percent area affected by drought for the state =72.8%

E. Staff position

Teaching

Cadre	No. of posts sanctioned	No. of posts filled
Maize Pathologist	2	2
Assistant Maize Breeder	1	-
Assistant Nutritionist	1	1

Non-Teaching

Cadre	No. of posts sanctioned	No. of posts filled
Senior Technical Officer	3	3
Field man	2	2
Lab assistant	2	1
Bullock man	1	-

F. Budget utilization for the year 2023-24

Regular Budget	Sanctioned (Rs.)	Released (Rs.)	Utilized (%)
ICAR Share 75%	1,18,31,375	80,47,375	100%
State Share 25%	39,43,792	26,82,458	100%
Total	1,57,75,167	**1,07,29,833**	

IIMR testing fee	Sanctioned (Rs.)	Released (Rs.)	Utilized (Rs.)	Utilized (%)
Kharif 2023	1,50,000	1,50,000	1,49,660	100%
Rabi 2023-24	90,000	90,000	88,031	100%
Total	2,40,000	2,40,000	2,37,691	**100%**

IIMR- FLD	Sanctioned (Rs.)	Released (Rs.)	Utilized (Rs.)	Utilized (%)
Kharif -2023	4,80,000	4,80,000	4,75,382	99.0
Area conducted 50 hectare in 125 farmers field.				

IIMR- SCSP	Sanctioned (Rs.)	Released (Rs.)	Utilized (Rs.)	Utilized (%)
Rabi 2023-24	4,00,000	4,00,000	2,27,550	56.9
Area conducted 20 hectare in 50 farmers field.				